

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0973

West Waynesboro 138 kV
July 04, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0973. The Interconnection Customer proposes a 108.8 MW solar photovoltaic facility interconnecting at the West Waynesboro 138 kV substation in the Pennsylvania area. The request was received on March 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,802,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Altavista Power Sta – Catoctin 138.0 kV
Contingency: W Kelso – Dry Gap 138.0 kV
Loading: 126.0% (Rating: 260 MVA)
DFAX: 6.4%
Pre-project loading: 123.3%
2. Facility: Dry Gap – Thurmont 138.0 kV
Contingency: Dry Gap – West Kelso Substation 138.0 kV
Loading: 124.7% (Rating: 459 MVA)
DFAX: 11.3%
Pre-project loading: 122.1%
3. Facility: Thurmont – Catoctin 138.0 kV
Contingency: W Kelso – Dry Gap 138.0 kV L
Loading: 122.7% (Rating: 285 MVA)
DFAX: 6.4%
Pre-project loading: 120.2%
4. Facility: Pineville – W Waynesboro 345.0 kV
Contingency: Highway – West Waynesboro 138.0 kV
Loading: 105.8% (Rating: 1,227 MVA)
DFAX: 18.0%
Pre-project loading: 104.2%
5. Facility: Pineville – Antietam 138.0 kV
Contingency: W Waynesboro – Pineville 345.0 kV
Loading: 103.5% (Rating: 665 MVA)
DFAX: 20.2%
Pre-project loading: 100.2%

6. Facility: Antietam – Clawson 138.0 kV
Contingency: Pineville – W Waynesboro 345.0 kV
Loading: 102.6% (Rating: 705 MVA)
DFAX: 20.2%
Pre-project loading: 99.5%
7. Facility: Reid – Clawson 138.0 kV
Contingency: West Waynesboro Substation – Pineville 345.0 kV line
Loading: 101.8% (Rating: 746 MVA)
DFAX: 20.2%
Pre-project loading: 98.8%
8. Facility: Reid – Palogen 138.0 kV
Contingency: W Waynesboro – Pineville 345.0 kV
Loading: 101.6% (Rating: 764 MVA)
DFAX: 20.2%
Pre-project loading: 98.7%
9. Facility: Highway – Palogen 138.0 kV L
Contingency: West Waynesboro – Pineville Substation 345.0 kV
Loading: 101.1% (Rating: 792 MVA)
DFAX: 20.2%
Pre-project loading: 98.3%
10. Facility: Highway – W Waynesboro 138.0 kV
Contingency: W Waynesboro – Pineville 345.0 kV L
Loading: 101.0% (Rating: 810 MVA)
DFAX: 20.2%
Pre-project loading: 98.3%
11. Facility: Bon Aqua – Dry Gap 138.0 kV L
Contingency: West Waynesboro – Pineville 345.0 kV Line
Loading: 100.2% (Rating: 813 MVA)
DFAX: 14.7%
Pre-project loading: 98.2%
12. Facility: Bon Aqua – Carl Bailey Substation 138.0 kV Line
Contingency: Pineville – W Waynesboro 345.0 kV
Loading: 100.1% (Rating: 820 MVA)
DFAX: 14.7%
Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Altavista Power Sta – Catoctin 138.0 kV	Rebuild line with higher-capacity conductor	\$10,485,000	40% (\$4,136,000)
Dry Gap – Thurmont 138.0 kV	Rebuild line with higher-capacity conductor	\$5,319,000	6% (\$313,000)
Catoctin – Thurmont 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,205,000	40% (\$2,448,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Waynesboro – Pineville 345.0 kV	Reconductor line	\$11,895,000	2% (\$273,000)
Pineville – Antietam 138.0 kV	Reconductor line	\$1,459,000	5% (\$68,000)
Antietam – Clawson 138.0 kV	Reconductor line	\$1,045,000	5% (\$49,000)
Reid – Clawson 138.0 kV	Reconductor line	\$1,590,000	5% (\$74,000)
Reid – Palogen 138.0 kV	Reconductor line	\$257,000	5% (\$12,000)
Highway – Palogen 138.0 kV L	Reconductor line	\$1,759,000	5% (\$82,000)
W Waynesboro – Highway 138.0 kV	Reconductor line	\$450,000	5% (\$21,000)
Bon Aqua – Dry Gap 138.0 kV	Reconductor line	\$2,263,000	7% (\$150,000)
Carl Bailey – Bon Aqua 138.0 kV Line	Reconductor line	\$2,659,000	7% (\$176,000)
	Total allocated to project		\$7,802,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.